

1. PROJECT OVERVIEW

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models

2. INTRODUCTION :

To overcome the limitations of manual illustration and the disconnected nature of standard AI art tools, our team introduces **ComicCraft**. This web-based platform serves as an autonomous digital comic studio, seamlessly merging Large Language Models (LLMs) with advanced Image Synthesis AI to automate the entire comic creation workflow.

3. SOLUTION HIGHLIGHTS:

Developed on the high-speed FastAPI backend framework, the solution offers a completely frictionless experience. Users interact with a clean, straightforward HTML interface where they define their narrative (Prompt, Main Character, Art Style). Once submitted, the system's background logic takes over—writing the dialogue, rendering the visuals, displaying a web preview, and compiling an offline document.

4. CORE SYSTEM FUNCTIONALITIES

- **Simplified User Interface:** A Jinja2-driven frontend that completely abstracts the complex API orchestration and prompt engineering from the end-user.
- **Intelligent Story Sequencing:** Deploys a dual-model approach using Google Gemini. One model handles the structural logic (JSON outlines), while the other focuses on creative prose (dialogues and narrations).
- **Autonomous Illustration:** Automatically translates the AI-generated scenes into high-quality API requests to Hugging Face Stable Diffusion, generating custom artwork on the fly.
- **Dynamic Web Display:** Generates an interactive preview page that visually loops through the rendered text and images directly in the browser.

- **Programmatic PDF Compilation:** Features a robust one-click export mechanism that uses the FPDF library to calculate page layouts and bind the story into a downloadable file.

5. THE USER WORKFLOW

1. The user visits the ComicCraft portal.
2. The user inputs their story parameters via dropdowns and text fields.
3. The server securely orchestrates the AI models in the background.
4. The generated comic is presented on the web interface.
5. The user downloads the final PDF document to their local machine.

6. VALUE PROPOSITION:

The proposed system drastically shrinks the comic production lifecycle from months to mere seconds. It is highly scalable, supports diverse artistic genres, and empowers anyone to become a published visual storyteller regardless of their technical background.